# PAPER_782: NGC 1672 — UQFF Barred Spiral Active Star Formation

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**Framework:** UQFF (Universal Quantum Field Superconductive Framework)
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## Abstract

NGC 1672 is a large barred spiral galaxy (SBb) ~60 million light-years away ($z \approx 0.004$) in the constellation Dorado, noted for its remarkably well-defined bar structure and prominent spiral arms containing numerous HII regions and star clusters. It was imaged by Hubble ACS in 2005 and again by JWST in 2023 with extraordinary resolution. NGC 1672 hosts an active nucleus and an above-average star-formation rate for its mass class (~3 M☉/yr), with its bar efficiently funneling gas toward the central starburst ring. Under UQFF, the elevated bar-driven SFR increases M_sf and the outflow velocity to $v = 2\times10^5$ m/s, doubling the standard result to g_NGC1672 $\approx 2.107\times10^{-3}$ m/s$^2$.

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## 1. Introduction

NGC 1672's bar is classified as one of the strongest (type SB rather than SAB), indicating efficient gas inflow to the central region. JWST NIRCam and MIRI imaging in 2023 revealed hundreds of star clusters in the spiral arms and the details of the central ring starburst. The combined bar-driven gas flow plus centrally concentrated starburst ring make NGC 1672 a higher-velocity system than symmetric spirals like M74. UQFF captures the bar-driven enhancement through $v = 2\times10^5$ m/s (double the symmetric spiral value) and M_sf = 0.06, yielding g_NGC1672 = $2.107\times10^{-3}$ m/s$^2$ — twice the standard result.

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## 2. Master UQFF Gravity Equation

$$g\_NGC1672(r, t) = (G \times M) / r^2 \times (1 + H(z)\times t) \times (1 + M\_sf) \times (1 + f\_TRZ) + a\_EM$$

### 2.1 Parameters

| Parameter | Symbol | Value | Source |
|---|---|---|---|
| Galaxy mass | M | $10^{11}$ M☉ = $1.989\times10^{41}$ kg | HST/JWST |
| Disk radius | r | $3\times10^{20}$ m (~32 kly) | NED |
| SFR (bar-driven) | — | 3 M☉/yr | JWST 2023 |
| Age | t | $5\times10^9$ yr = $1.578\times10^{17}$ s | Hubble time |
| M_sf | — | 0.06 | UQFF bar-enhanced |
| Redshift | z | 0.004 | Spectroscopic |
| v_EM | v | $2\times10^5$ m/s | Bar-driven outflow |
| B_EM | B | $10^{-5}$ T | Galactic field |
| f_TRZ | — | 0.05 | UQFF bar |

## 3. Long-Form Derivation

### Step 1: Base Gravitational Term

$g\_grav = 6.6743e-11 \times 1.989e41 / (3e20)^2 = 1.476e-10\ m/s^2$

### Step 2: Cosmic Expansion Factor

$H(z) = 2.285e-18\ s^{-1}$; $H(z)\times t = 0.361$; factor = 1.361

### Step 3: SFR Mass Fraction (Bar-Enhanced)

$M\_sf = 0.06$; $1 + M\_sf = 1.06$

### Step 4: Time-Reversal Correction

$f\_TRZ = 0.05$; $1 + f\_TRZ = 1.05$

### Step 5: Gravitational Total

$g\_grav\_total = 1.476e-10 \times 1.361 \times 1.06 \times 1.05 = 2.237e-10\ m/s^2$

### Step 6: Aether EM Correction (Bar-Enhanced v)

$a\_EM = (1.602e-19 \times 2e5 \times 1e-5 / 1.673e-27) \times 11 \times 1e-12 = 2.107e-3\ m/s^2$

### Step 7: Final Solution

$g\_NGC1672 = 2.237e-10 + 2.107e-3 \approx 2.107e-3\ m/s^2$

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## 4. Physical Interpretation

NGC 1672's strong SB bar drives gas flow at double the symmetric spiral velocity. JWST 2023 imagery confirms the central starburst ring fed by this bar, validating $v = 2\times10^5$ m/s as the UQFF bar-flow parameter. The 2× enhancement relative to standard spirals ($g = 2.107\times10^{-3}$ vs. $1.053\times10^{-3}$ m/s$^2$) directly reflects the bar-driven kinematic intensification captured by UQFF's linear v-dependence.

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## 5. Conclusions

UQFF applied to NGC 1672 yields $g \approx 2.107\times10^{-3}$ m/s$^2$, exactly double the standard SBbc result. Strong bar-driven gas inflow and the central starburst ring, confirmed by JWST 2023, validate $v = 2\times10^5$ m/s as UQFF's bar-drive velocity parameter.

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